Neurals

A Rarity System for Neural Networks

neurals.network

Abstract

A trained language model is more than a billion numbers arranged in layers, yet once it is published it becomes a file anyone can copy, with no record of who stood behind it and no way to hold a stake in any part of it. We propose a rarity system for neural networks. An open source model, Qwen2.5 1.5B Instruct, is written byte for byte onto Solana so that it can be rebuilt from the chain alone. Its anatomy is divided into 1,318 real parts, and every one of its 250,880 MLP neurons is owned through a Solana token at 3,000 tokens per neuron. Rarity is measured, not drawn. Five traits found by interpretability research are located in the inscribed file before launch and committed by hash, and a random reveal decides which part each buyer receives. Every paid message produces a signed receipt of how hard each part fired, and the fee is split among the owners of the parts that did the work. The network is drawn as a live three dimensional model in which every part can be found and watched. Neurals is an experiment.

1. Introduction

Training a capable model costs millions of dollars, and once its weights are released that investment becomes a file that can be copied without limit. Open weights have made modern AI available to anyone, but they have left the models themselves without provenance and without any way for the people who value them to hold a stake in them.

Today's open models are a first generation, and most will be replaced or forgotten within a few years. What is needed is a way to freeze one exactly as it was, divide it into its real parts, and record permanently who holds each one. Ownership here is not control over the model or its license. It is a public, verifiable record that a wallet holds a particular part of a particular model, together with a share of the fees the model earns when that part is used.

2. The Model

The model is Qwen2.5 1.5B Instruct, an open source model released under an open license [3]. It has 28 layers, a hidden size of 1,536, 12 attention heads and 8,960 MLP neurons per layer, and a vocabulary of 151,936 tokens. It was chosen as the first experiment because it is small enough to inscribe in full and open enough to measure without restriction.

The exact file people own is the file that is measured, a 4 bit GGUF file of about 1.1 GB. It is written to Solana in 960 byte chunks, roughly 1.15 million transactions and about 6 SOL in fees, with a manifest of SHA 256 hashes. The runner, llama.cpp compiled to WebAssembly, is written too, so the model can be rebuilt and run from the chain alone.

3. Parts

The unit of ownership is the neuron, and every 3,000 tokens bought is one MLP neuron. Parts are real pieces of the model, so they differ in size, and their total equals the token's saleable supply of 793,100,000 exactly.

PART	COUNT	WHAT IT IS	TOKENS EACH
Embedding	1	Where text enters the model	3,430,000
Attention head	336	One head, 28 layers of 12	100,000
Neuron column	980	256 MLP neurons, 28 layers of 35	768,000
Unembedding	1	The output head, where the next token is chosen	3,430,000

Each part belongs to one of six stages by its layer, following what research broadly finds each depth of a model doing [10, 11]. Tokenizer layers 0 to 3 join word pieces into words, syntax layers 4 to 7 handle grammar, semantic layers 8 to 13 build meaning, memory layers 14 to 19 hold stored facts, reasoning layers 20 to 24 combine them, and prediction layers 25 to 27 choose the next token.

4. Buying and the Reveal

Position 0 is the first token bought, and positions fill slots in a fixed order. The first slot is always the Embedding and the last is always the Unembedding. Between them, each of the 28 layers holds 12 head slots followed by 35 column slots. A buyer knows the kind of part they bought but not which one.

After the curve sells out or reaches its deadline, a seed block that nobody can choose draws a permutation assigning the real heads and columns to their slots. The traits of every real part are public before launch while the mapping is not, so nobody can aim a buy at a rare part. Anyone can recompute the reveal in their browser.

5. Rarity

Traits are measured facts about real parts of the inscribed file, found before launch by a published tool and a committed reference text. Luck lies in which part a buyer receives, not in what the parts are.

TIER	TRAIT	WHAT IT IS	PARTS
★★★★★	Super Weight	Holds one of the model's largest outlier weights [4]	3
★★★★☆	Induction Head	Spots a pattern earlier in the text and continues it [6]	6
★★★☆☆	Massive Activation	Fires thousands of times harder than typical [5]	6
★★☆☆☆	Attention Sink	Parks attention on the first token to keep the model stable [7]	3
★☆☆☆☆	Monosemantic	Responds to one clear thing rather than many [9]	24
☆☆☆☆☆	No trait	Earns by how hard it fires alone	1,276

A part with a trait worth p points counts in the fee split with weight

$$w = (100 + p) / 100$$

so a Super Weight counts 2.0 times and a Monosemantic part 1.3 times. Each part takes at most one trait, and once committed the thresholds and results never change.

6. Commitment and Verification

Before the token exists, a memo is posted on chain holding the SHA 256 hash of the rules file, which contains the slot layout, the measured traits of every part, the hashes of the tool and reference text, and the seed rule. Anyone can rerun the measurement and obtain the same list, recompute the reveal, rebuild the ledger of positions from the chain, and replay any receipt to within one part per million.

7. Holding

Neurons never die. A wallet's neurons are awake as far as its token balance covers them, and when the balance falls the rarest stay awake first while the rest go dormant. Dormant neurons earn nothing, and buying back wakes them. Fees use the lowest balance of each hourly epoch, so buying just before a payout earns nothing.

To trade neurons without moving tokens, an owner seals them into a checkpoint, a program controlled account that freezes neurons together with their tokens. A checkpoint can be listed, bought, gifted or withdrawn, and the market fee is 1.5 percent.

8. Receipts and Fees

Each paid message costs 0.001 SOL. After the reply, the prompt and reply are run again in one canonical pass using the WebAssembly build, whose arithmetic is identical on every machine, and each part's activation mass is summed. Each kind of part receives a fixed budget B_k in proportion to the tokens it was sold for, and within it a part's share is

$$s_i = B_k \, w_i \, m_i \, / \, \Sigma_{j \in k} \, w_j \, m_j$$

where m_i is its activation mass. The receipt, holding the model hash, the tokens and all 1,318 shares, is signed by the model's key and posted with the fee. Each hour, owners are paid for every position held awake for the whole epoch. The token gives a share of usage fees to holders of awake neurons, and no control over the model, its license or the project.

9. Drawing the Network

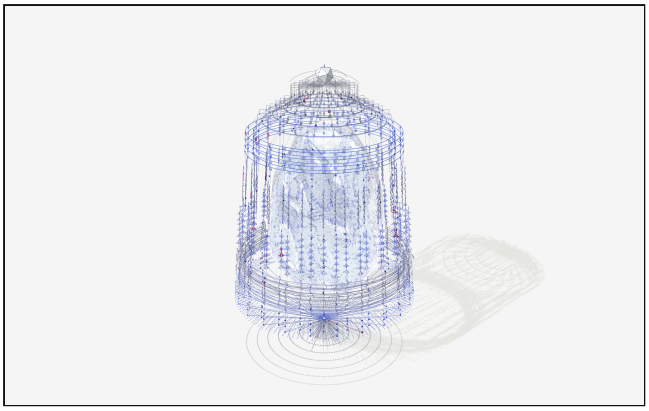


Figure 1. The network, data in at the top and data out at the bottom.

Models represent concepts as shapes, days as circles and quantities as curves [12, 13]. Neurals applies the same idea to the architecture, so that every property of the drawing is assigned to one quantity in the model.

IN THE DRAWING	IN THE MODEL
Height	Depth, layer 0 at the top to layer 27 at the bottom
A ring	One layer and its 47 parts
Angle around a ring	Nothing. The parts in a layer can be reordered without changing the output, so a circle, with no first or last place, is the honest shape
Solid at the top	The Embedding, where text enters
Core and floor rings at the bottom	The Unembedding, and words leaving the model
Brightness of a part	How hard it fired on real messages
Outer surface	Every ownable part, kept visible
Water at the center	The token's liquidity, the one element that is not the model

Each stage is drawn in its own way, circuit traces for the tokenizer, gate cells for syntax, hanging chains for meaning, branching neurons for memory, lightning for reasoning and a field of cells for prediction. The narrow crown, the starting angle of each ring and the slow stretch of the stages are stylistic. Everything else is data.

10. Icons and Color

TOKENIZER	A chip with a trace, pieces routed and joined
SYNTAX	A gate cell, one rule in a chain of rules
SEMANTIC	A hanging chain, meaning built link by link
MEMORY	A neuron with dendrites, where knowledge is held
REASONING	A forked filament, facts meeting and branching
PREDICTION	A target over a row of candidate tokens
EMBEDDING	An arrow entering, text coming in
ATTENTION HEAD	A ringed nucleus, deciding where to focus
NEURON COLUMN	A string of beads, a stack of neurons
UNEMBEDDING	An arrow leaving, words going out

The network is drawn in two tones, navy and cobalt, alternating by stage. Ownership has three colors, and pink is used for nothing else.



Figure 2. The three ownership colors.

11. Liquidity

The water inside the network stands for the token's liquidity pool. Its size follows the market cap M, read from public market data once a minute, with the full drawn size standing for 1 million dollars. Because the water is a volume, its radius scale is

$$r = (M / 10^6)^{1/3}$$

so a market cap eight times larger makes the water twice as wide. It never shrinks below 22 percent, so it is visible from launch.

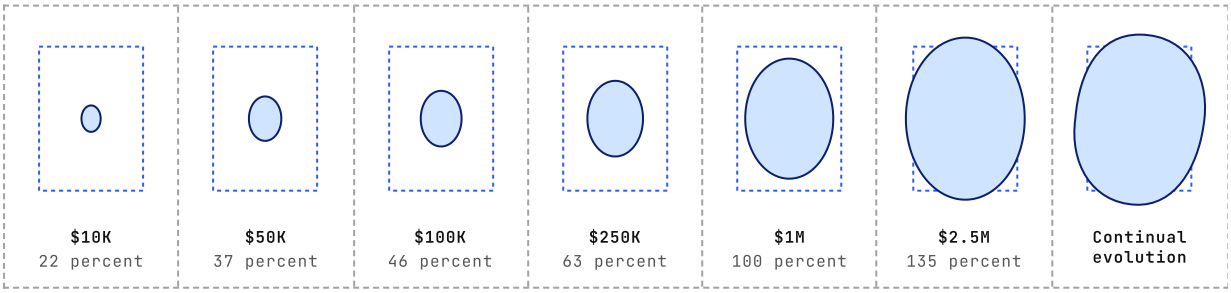


Figure 3. Water size by market cap.

12. The Simulator

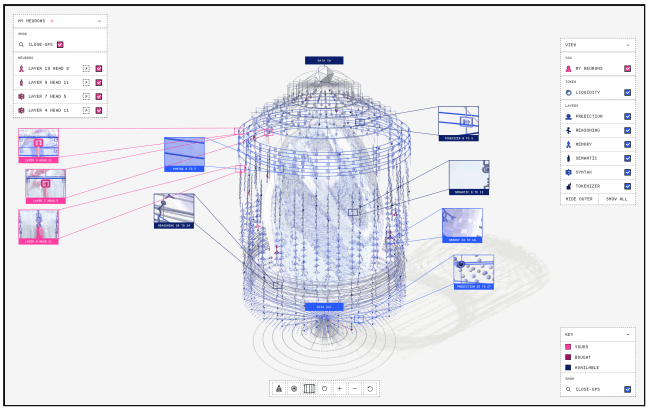


Figure 4. The network with its boxes and bottom bar.

The network is a live simulation built so that a holder can find any part in seconds. It turns slowly on its own, dragging turns it and scrolling zooms. The bottom bar gives four views, full, birds eye, up close and inside, with stepped zoom and a restore button. The View box shows or hides each stage and the water, and peels stages away to reach parts deep in the stack. The My neurons box lists every part you hold with a button that flies the camera to it. Live close ups magnify each stage like an anatomy textbook. Every part on the site links into the network at its exact place, and a guided tour flies through all six stages in the order the model computes.

13. Calculations

Saleable supply	793,100,000 tokens
Tokens per neuron	3,000
Neurons owned	250,880
Ownable parts	1,318
Parts with a trait	42
Average share per part	about 760 parts per million
Inscription	about 1.15 million transactions, 6 SOL
Message price	0.001 SOL
Epoch	one hour

14. Security and Limitations

Gaming the measurement is prevented by committing the method before the token exists, and aiming at rare parts by a reveal nobody can choose. Made up receipts are rejected because only receipts signed by the model's key count, and a posted payout waits out a challenge window during which a multisig can veto it.

Ownership is provenance and a share of fees, not control. Traits depend on the method used to measure them. The Super Weight trait was redefined after measurement found no single weight in this compressed model breaks it when removed, so it now names the largest outlier weights. The archive is as permanent as the chain, and legal treatment varies by country.

15. Beyond the First Model

The method is designed to carry over, and the team is working toward two further steps. The first is applying the same rarity system to other open source networks, each inscribed, measured and committed under the same rules. The second is investigating whether it can extend to closed models, whose weights are not public. That would require a different form of commitment and verification, and it remains an open question rather than a promise.

16. Conclusion

We have proposed a way to own the parts of a neural network without closing it. The model is written to the chain, divided into its real anatomy, measured, committed and sold so that nobody can aim at rarity. Every paid message leaves a verifiable record of which parts did the work, and the fee follows that record. The result is a permanent, part by part record of an early model, and a way to see it.

References

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